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| **COMP1801** | **Machine Learning** | **Faculty Header ID:** | **Contribution: 100% of course** |
| **Module Leader: Dr Ilya Alexakhin** | **COMP1801 Coursework** | | **Deadline Date: Wednesday 09/12/2026 (17:00 UK time)** |

This coursework should take an average student who is up-to-date with tutorial work approximately 50 hours.

Feedback and grades are normally made available within 17 working days of the coursework deadline. Please be aware that this does not include any bank holidays or Christmas/New Year's break. The official return deadline is thus 9th of January, 2027. If needed, any changes to the return deadline will be announced via Moodle/Outlook.

**Learning Outcomes:**

1. Demonstrate knowledge and understanding of the major approaches to machine learning.
2. Understand the different types of learning algorithms.
3. Critically evaluate the merits and limitations of machine learning approaches and algorithms and evaluate the choice of algorithms for specific real-world contexts and requirements.
4. Apply machine learning algorithms to selected real-word problems in practice and understand the processes involved in their deployment.

**Plagiarism and Use of Generative AI in this Assessment**

**Plagiarism** is presenting somebody else's work as your own. It includes: copying information directly from the Web or books without referencing the material; submitting joint coursework as an individual effort; copying another student's coursework; stealing coursework from another student and submitting it as your own work. Copying from your previously produced work is also considered to be a form of plagiarism, or self-plagiarism. Suspected plagiarism will be investigated and if found to have occurred will be dealt with according to the procedures set down by the University. Please see your student handbook for further details of what is / isn't plagiarism.

**All material copied or amended from any source (e.g. internet, books) must be referenced correctly according to the reference style you are using.**

**Generative AI may be used only to support preparation for this assessment.**

Permitted uses include learning support, planning, brainstorming, clarification, comprehension, feedback, and language improvement. AI must not generate substantive assessment content.

All analysis, arguments, calculations, interpretations, and the final submission must be your own work.

**Submitting AI-generated work as your own will be treated as assessment misconduct. Your work will be submitted for plagiarism checking.**
**Any attempt to bypass our plagiarism detection systems will be treated as a severe Assessment Offence.**

## Coursework Submission Requirements

An electronic copy of your work for this coursework must be fully uploaded by 17:00 (UK Time) on the Deadline Date of **Wednesday 09/12/2025** using the link on the coursework Moodle page for COMP1801.
For this coursework you must submit **2 separate files** (separate submission portal links on Moodle for each file):

- a single PDF document containing your final report.
- a ZIP file containing your code: both as a IPYNB workbook and as a PDF exported from collab. (.PY files are not allowed as they lack outputs)
    - Its OK if the PDF export of your code looks strange or has improper formatting, it is just to check against the IPYNB file as an anti-cheat measure.

Please note:

- Any text in the document must not be an image (i.e. must not be scanned) and would normally be generated from other documents (e.g. MS Office using "Save As… PDF"). An exception to this is handwritten mathematical notation, but when scanning do ensure the file size is not excessive. Take care to not “flatten” the PDF during exportation and saving, in order to avoid text being merged into a single image.
- There are limits on the file size (see the relevant course Moodle page).
- Make sure that any files you upload are virus-free and not protected by a password or corrupted otherwise they will be treated as null submissions.
- You must NOT submit a paper copy of this coursework.
- All course-works must be submitted as outlined above. Under no circumstances can they be accepted by academic staff by other means (email, Teams, etc.).
- DO NOT send a draft of your coursework to any members of teaching staff for feedback and review prior to submission. We (staff) are barred by policy to give detailed feedback or make any direct contributions to your work (like said feedback). If you would like help or some generic feedback on specific parts of your CW draft, please approach us (staff) directly in the labs.

## Academic Misconduct and Extenuating Circumstances

Please refer to the following links for the University’s Academic Misconduct and Extenuating Circumstances policies and procedures:

- Assessment Misconduct Procedure (Taught Awards) | Documents | University of Greenwich

- Extenuating circumstances | Student Services | University of Greenwich

If you still have any strictly admin-related questions and/or queries, please contact FES Student advisors fes-studentadvice@greenwich.ac.uk for further advice.

# Detailed Specification

***This Coursework is to be completed individually.***

You are working as a Data Scientist for a (fictitious) manufacturing company which produces metal parts for various industries. They have been experimenting with a new metal alloy that should have superior properties to the one being currently used. However, in practice it was found to be very sensitive to changes in the processing parameters during production, causing defects to form. If these defects reach a certain size or quantity, they severely impact the lifespan of the metal parts, meaning the entire part must be scrapped as defective.

As the process of measuring a parts lifespan is a slow, destructive, and expensive process, the company hopes that you can create a Machine Learning solution that allows them to estimate if a part has sufficient lifespan to be considered "safe" for use. Based on more easily measured parameters, which the company hopes to use to refine their production process to lower the incidence of defects.

The company has collected a dataset consisting of a table of processing parameters and measurements taken from the completed parts. This provided set of data has been cleaned, so there is no need to check for incorrect, duplicate, or missing data.

## Data

To complete this assignment, you MUST use the data provided, found on this URL:

https://moodlecurrent.gre.ac.uk/mod/resource/view.php?id=4658069

It is also provided on Moodle along with more detailed descriptions of the dataset. It is recommended that you read these descriptions to aid in your understanding of the problem domain.

Submissions based on other data will not be marked and will receive 0 marks.

Note: This data was created specifically for this assessment and is entirely synthetic.

## MSc Standards

Remember that at the MSc level, the most important learning outcome (i.e. the thing that gets the highest mark) is **evaluation**, which is the examination of "why" something occurs, or reason a choice is made. Thus, ideally, any choice, method, result and/or discussion would include as much evaluation as possible in addition to the detailed description of the above (i.e. "what" and "how").

Therefore, attempt to always approach all the following requirements with the "what/how/why" mindset when writing MSc reports (course-works, MSc project and all academic writing in general from now onwards.) More on this will be discussed during the CW Workshop during week 7.

## Report

Your overall task is to fit a selection of Machine Learning methods to this data, evaluate them, and finally compare them to provide a recommendation for which specific method the company should apply to the rest of their data.

This must be presented in the form of a report to your line manager. This report must be split into the seven parts below (shown with the corresponding mark weighting), with marks also being rewarded for the presentation and language in your report.

Your report should contain images, tables, equations and/or other visualizations to help you demonstrate your work in parts 2-4. It is down to your discretion how many images and/or tables you wish to include, what they should show and how they are split across the different parts of the report.

### Part 1 – Executive Summary (5 Marks)

This should be a short, but concise summary of what the report contains – the problem you are solving, why it's important, the ML methods you have used and a summary of your results and conclusions. Avoid generic statements about machine learning, A.I., data processing etc., and keep to the specifics of the coursework.

Tip: Think of the Abstract section of academic papers for a rough idea.

### Part 2 – Data Exploration (10 Marks)

Describe how you loaded the dataset, then perform a concise exploration of the data and comment on any patterns or relationships in the data that you think may be relevant for creating a model and predicting the part lifespan. It is advisable to provide plots and visualizations to better highlight these regions of interest.

Tip: You can use the patterns/relationships shown in this section to better justify your chosen methods in later sections.

Considering this data exploration, identify the features you will use in your models and discuss why you believe they will be the best predictors of metal part longevity.

Finally, with reference to patterns/relationships found in the exploration and (if relevant) theoretical justification, provide a brief additional discussion outlining your expectations of which approach (i.e. regression or classification) and specific ML model will be most appropriate for providing an accurate solution to the company's requirements.

Tip: This will also be useful in the Conclusions section (**Part 5**), where your initial discussion here can be compared with your post-experimentation results.

## Part 3 – Regression Implementation (30 Marks)

### 3.1 Methodology (10 Marks)

The task for this section is to create a regression model methodology to predict the lifetime of a metal part using machine learning methods.

**Firstly**, choose two model types appropriate for this task, for example Linear Regression and Artificial Neural Network. Both models chosen should be supported with a brief justification explaining why it is an appropriate choice for the problem. Parametric models are encouraged (especially neural networks,) but not required for a passing grade. Decision Trees and Random Forests count as one method. Baseline models do not count towards your model choices.

Tip: The models chosen do not have to be one of the models taught in the lectures (though there is no penalty for only using taught models).

Tip: The justification for choosing a model could include (but is not limited to): Patterns/relationships found in **Part 2**, reference to other sources, some basic experimentation, logical reasoning.

**Secondly**, describe and implement an appropriate pre-processing routine to be used in all following experimentations and evaluations of this section only (**Part 3**). This can include, but is not limited to: categorical feature encoding, feature scaling, splitting the data into training and test sets, data balancing, etc.

**Thirdly**, outline the details of the experimental set-up. These could include (but not limited to): which performance metrics will be used and why, any baseline (or benchmark) against which model performance will be measured, any stopping criterion used to deem model performance as satisfactory, etc.

Tip: A flowchart (with brief explanations) helps to illustrate the workflow of this sub-section without adding (much) to the word count.

Tip: A single baseline is typically enough for comparison with both model types. This can be as naïve as using the mean value (for regression,) however, simple models are encouraged.

Lastly, provide and justify the hyper-parameter tuning framework you will use to obtain your final model versions in the experiments in section **3.2**. Describe which hyper-parameters you will tune for each model type chosen, how they work and why they are essential to the optimization of that architecture.

### 3.2 Evaluation (15 Marks)

For this sub-section you should perform and describe the experiments done to obtain your final regression model version for each chosen architecture type, by adhering to the hyper-parameter tuning framework outlined above. These experiments should feature a rigorous model optimization process including comparisons between the

different versions via a table or otherwise detailed description. All choices should be justified using theory, experiments and/or references.

The final model versions of both chosen types should be described in all detail – summarizing all relevant final hyperparameters chosen.

Reminder: Make sure that you evaluate your final model versions using the test portion of the dataset not used in the prior training stages, using appropriate regression performance metrics. You should explain how these chosen metrics work, why they are appropriate to the task and provide a written interpretation of how well your model is performing at the given task according to these metrics.

Finally, compare the best performing versions of both model types using your chosen performance metrics. Provide your final recommendation of a single model type's version for deployment on this regression task, supporting this choice with a brief discussion of your results for justification.

Tip: Remember to use the same train/test split ratio and seed for all model versions and types for a fair comparison. Also remember to use the same performance metrics across all models for parity.

Tip: A stronger evaluation section would generally include a table showing the hyper-parameter tuning progression to arrive at a best model version for both chosen model types (at least two such tables), followed by a direct comparison between the best versions of each model type (an additional table).

### 3.3 Critical Review (5 Marks)

For this sub-section critically review your overall methodology used for this task only (**Part 3**), considering the results obtained in your experiments in section **3.2**. Cover areas of strengths and areas where improvement might be needed. Offer an alternative approach from the choices not utilized in your experimentations that future investigations could explore. These can include untrialled model architectures, alternate pre-processing routines, different hyper-parameter tuning schemes, etc.

## Part 4 – Classification Implementation (40 Marks)

### 4.1 Feature Crafting (10 Marks)

Your line manager has decided that beyond the exact lifetime of a part, it is also important to know whether a part has a lifetime above 1500 hours (determined by the company as the minimum lifetime before a part is considered defective).

Before choosing any potential model architectures, first create an additional feature (a column named "1500_labels" for example) representing a binary output label for this lifespan threshold that may be used to predict whether a part is defective or not. Populate this feature with a positive binary output if the hourly threshold is met.

However, your line manager thinks that splitting the data into only two groups may be naïve, and hence predictions made by a binary classification model may not be suitable for finding the best processing parameters for manufacture. Currently, it is unclear how many groups the data should be split into and why.

Instead of using the provided threshold of 1500 to create a binary class, you can (**for higher marks**) perform and utilize alternate grouping methods on the lifespan feature (and potentially relationships with the other features,) while providing justifications for doing so. The type of this grouping can range from a more complex thresholding technique (such as a statistical outlier function) to the application of an unsupervised learning algorithm, such as clustering, to separate the inputs into different groups (which can result in three or more groups.) More complex and/or better-justified methods will earn higher marks for this section than simpler ones, but must be researched by the student on their own.

Important note: If you are choosing to utilize an alternate threshold grouping, the subsequent classification methodology and evaluation sections must utilize these groupings in all following experiments and discussions of **Part 4**! This can change your prediction task into a multi-class or multi-label classification. If these sections are found to be using the provided binary threshold of 1500 instead, higher marks will not be awarded for all following/relevant sections.

Tip: If using a clustering method, remember you need to choose a value for the number of clusters yourself (**k** in K-means for example) and justify this value with appropriate reasoning and/or experimentation. References can help here.

You should include at least one plot or table showing the results of the thresholding or grouping to give a better idea of how balanced the resulting dataset is. If using thresholding/grouping other than 1500, provide justifications as to why it is superior.

### 4.2 Methodology (10 Marks)

The task for this section is to create a classification model methodology to predict the output class of a metal part using machine learning methods.

Firstly, once your output labels are created, choose two model types appropriate for this task, for example Logistic Regression and Artificial Neural Network. Both models chosen should be supported with a brief justification explaining why it is an appropriate choice for the problem. Parametric models are encouraged (especially neural networks,) but not required for a passing grade. Decision Trees and Random Forests count as one method. Baseline models do not count towards your model choices.

Tip: The models chosen do not have to be one of the models taught in the lectures (though there is no penalty for only using taught models).

Tip: The justification for choosing a model could include (but is not limited to): Patterns/relationships found in **Part 2**, reference to other sources, some basic experimentation, logical reasoning.

Secondly, describe and implement an appropriate pre-processing routine to be used in all following experimentations and evaluations of this section only (**Part 4**). This can include, but is not limited to: categorical feature encoding, feature scaling and splitting the data into training and test sets, data balancing, etc.

Tip: Remember to exclude the lifespan feature from both train and test set inputs for this task, since we are trying to predict its threshold/groups from the other variables using our newly created output labels (which is trivial if we keep lifespan as an input feature).

Thirdly, outline the details of the experimental set-up. These could include (but not limited to): which performance metrics will be used and why, any baseline (or benchmark) against which model performance will be measured, any stopping criterion used to deem model performance as satisfactory, etc.

Tip: A single baseline is typically enough for comparison with both model types. This can be as naïve as using a "random draw" (for classification,) however, simple models are encouraged.

Lastly, provide and justify the hyper-parameter tuning framework you will use to obtain your final model versions in the experiments in section **4.3**. Describe which hyper-parameters you will tune for each model type chosen, how they work and why they are essential to the optimization of that architecture.

Reminder: All choices of models must be capable of and configured to perform the same task (e.g. binary **or** multi-class) to ensure parity. High grades are still possible if using only binary classification. Implementing a multi-class classification model is for those seeking a challenge for potentially top marks. Hence, if you are constrained for time, or are otherwise having difficulty with this section, it is recommended to use the 1500 thresholding rule for a binary classification problem and focus on providing the best discussion and evaluation possible.

### 4.3 Evaluation (15 Marks)

For this sub-section you should perform and describe the experiments done to obtain your final classification model version for each chosen model type, by adhering to the hyper-parameter tuning framework outlined above. These experiments should feature a rigorous model optimization process including comparisons between the different versions via a table or an otherwise detailed description. All choices should be justified using theory, experiments and/or references.

The final model versions of both chosen types should be described in all detail – summarizing all relevant final hyperparameters chosen.

Reminder: Make sure that you evaluate your final model versions using the test portion of the dataset not used in the prior training stages, using appropriate classification performance metrics. You should explain how these chosen metrics work, why they are appropriate to the task and provide a written interpretation of how well your model is performing at the given task according to these metrics.

Finally, compare the best performing versions of both model types using your chosen performance metrics. Provide your final recommendation of a single model type's version for deployment on this classification task, supporting this choice with a brief discussion of your results for justification.

Tip: Remember to use the same train/test split ratio and seed for all model versions and types for a fair comparison. Also remember to use the same performance metrics across all models for parity.

Tip: A stronger evaluation section would generally include a table showing the hyper-parameter tuning progression to arrive at a best model version for both chosen model types (at least two such tables), followed by a direct comparison between the best versions of each model type (an additional table).

### 4.4 Critical Review (5 Marks)

For this sub-section critically review your overall methodology used for this task only (**Part 4**) considering the results obtained in your experiments. Cover areas of strengths and areas where improvement might be needed. Offer an alternative approach for future investigations to explore from the choices not utilized in your experimentations. These can include untrialled model architectures, alternate pre-processing routines, different hyper-parameter tuning schemes, etc.

## Part 5 – Conclusions (5 Marks)

Summarize your findings in-light of your experiments' results and compare them to your initial assessment (from Data Exploration section, **Part 2**). Attempt to provide a final clear recommendation to your manager for which single model version should be deployed between the Regression (**Part 3**) or Binary Classification (**Part 4**) implementations for the task in helping the company predict if a part is usable or not. Provide a reasoned justification for this recommendation considering aspects such as (but not limited to) model metrics, feature crafting and the outside business context of the task (while keeping it relevant to machine learning). If you are not comfortable recommending either model be deployed, explain why.

## Part 6 - Report Presentation, Language and References (10 Marks)

The report should be presented in a professional manner with a neat and clear layout and all writing in proper English using good grammar.

The report must be written using one of the following templates (make your own copy to work on):

- Word – Find on the course Moodle page (direct link)
- Latex – Find on the course Moodle page (direct link)

To obtain full marks your report should adhere to the following:

- Follow the template page layout (in either latex or word):
  - No egregious changes to font type, size or colour.
    - **12 Arial** for Word in fine
  - No changing the page layout (e.g. margin, orientation, size)
  - Text should be split into sections with titles and proper paragraphs
- All images and tables should have a label and will be ignored unless specifically referred to using this label in the body text.
- Images should be properly generated, labelled, and rendered at an appropriate scale and resolution.
  - Images are expected to primarily be graphs, plots and other visualizations that can be used to aid in describing and understanding your data and the model selection and evaluation processes used.
  - If you wish to include images of the loaded data, this should only be an indicative sample showing up to 10 rows.
  - **You should not include screenshots of your code in the report.**
- Total word count should not go over 3000 words. There is no minimum word count, but it is recommended the report should have at least 2000 words.
- The individual tasks do not have word counts assigned. It is down to your discretion how much space to devote to a section given the brief and the number of marks provided.
- Correct spelling, punctuation and grammar should be used in all text.
- You need to include a reference list that should include any external references used such as academic papers, books, blogs, and code repositories. While explicit marks are not awarded for the reference list, to achieve good marks you would be expected to use multiple references throughout the report in each section to support your decision making and interpretations of the results. In text citations should be used and all references should use a consistent referencing style. Note there should not be any references in the executive summary. Reference section does not count towards the total word count.
- You may include a table of contents and list of figures/tables if you like, but try to keep it neat and nicely formatted.
- You do not need to include a cover page, but will not be penalized if you do.
- Marks may also be removed at markers discretion for other issues with presentation not listed above.

## Python Code

You should implement a solution to the above task within a single Interactive Python Notebook (IPYNB) using Google Colab. Any data in the text/tables or images used in your report should be generated using this Python Notebook. Please use IPYNB file format and not PY in order to preserve printed outputs and any markdown text.

This source code must also be provided as both a downloaded notebook (IPYNB) and exported as a PDF from Google Colab.

Markers need to be able to run your provided code without any errors requiring debugging. Ideally, we (markers) should be able to hit "run all" in Colab and have the notebook run start to finish without error. Because of this, we strongly encourage you to use Colab at least in the final run-through of the notebook prior to the preparing the final export of the IPYNB file (and PDF copy) for submission.

If the markers are unable to reproduce any results in your report, you may be penalized for the inconsistency. Please make an attempt to keep your notebook as neat and organized as possible as well, it really helps the marking process overall!

Remember: if a marker cannot find something in your code (and report) due to it being vague or made difficult to spot, it is not their responsibility to dig through the code/text to find it. Attempt to make it as obvious as possible for the marker to find the appropriate section(s) in order to make awarding marks as easy as possible! The official term for this is called "signposting," so please attempt to signpost as much as possible.

However, the **markers will not be marking your source code** and will only look at it in detail if they feel they need to test or otherwise check something you present in the report. Your marks are based on the REPORT ALONE and **not** the code, so ensure that everything required to complete the tasks given are shown in the report. Even if your code provides a good solution, anything not shown/explained in the report will not be marked.

## Assessment Criteria

### Marks Breakdown

The mark breakdown is given below. Other than the report presentation, the marks given for each section are typically somewhat proportional to the amount of space you should devote to a section in the report. Generally, both the quality of the solution implemented and the explanation of your reasoning and interpretation in the report will be considered.

- Executive Summary **(5 Marks)**
- Data Exploration **(10 Marks)**
- Regression Implementation **(30 Marks)**
  - Methodology (10 Marks)
  - Evaluation (15 Marks)
  - Critical Review (5 Marks)
- Classification Implementation **(40 Marks)**
  - Feature Crafting (10 Marks)
  - Methodology (10 Marks)
  - Evaluation (15 Marks)
  - Critical Review (5 Marks)
- Conclusions and Recommendations **(5 Marks)**
- Presentation and language **(10 Marks)**

### Rubric

Some general additional guidelines you should consider on how marks are awarded:

- Model accuracy: The final accuracy/loss of your ML models do not have a direct impact on your marks! While you are expected to experiment and obtain the best accuracy that you can, it is recognised that this is not always an easy task. Rather, the mark will mostly depend on your overall approach depth to the problem, the soundness of your reasoning, and how you evaluate and analyse your results (the "what, how and why.")
- Model complexity: While we are not explicitly marking you higher for implementing more complicated models in your solutions, higher scores will often correlate with more complex implementations as there is generally more scope to experiment with and discuss their hyperparameter tuning (so long as they are well discussed and justified as usual.) For example, a simple multivariable linear regression model may provide a valid solution with reasonable accuracy, but there are no hyperparameters to tune in this case and hence nothing to experiment with for improving the model accuracy. This would inevitably lead to some marks not being awarded in the methodology and evaluation section for this model.
- Explanation detail: Demonstration of a critical understanding of the relevant concepts, appropriate explanation of all your steps taken and recommendations/discussions that are supported by your results.
- Report quality: Are all sections included and properly completed? Does it read well? Does each section have a logical structure? Are design decisions appropriately discussed? Is your evaluation realistic and thorough?

Please see table below for the full rubric of how marks will be allocated:

| | 0-29% Fail | 30-49% Fail | 50-59% Good | 60-69% Very Good | 70-79% Excellent | 80-100% Exceptional |
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| **Executive Summary (5 Marks)** | A completely unsatisfactory or missing summary of the work done which fails to cover almost any of the basic points. | A poor summary of the work done which fails to cover many basic requirements. | A good summary of the work done covering most of the requirements. | A very good summary of the assignment covering all relevant tasks in detail. | An excellent summary of the assignment lacking only minor details. | An outstanding summary with very little that could be improved. |
| **Data Exploration (10 Marks)** | Absent or very little evidence of any exploration of the data. Data loading not mentioned. | A poor exploration of the data that is largely incomplete or containing mistakes. Implications are scarcely mentioned, and the data loading is barely described. | A good exploration of the data that may be incomplete or contain mistakes. Some implications are discussed, and the data loading is adequately described. | A very good exploration of the data that is mostly complete but may contain mistakes. Implications are discussed clearly, and the data loading is clearly described. | An excellent exploration of the data that is comprehensive with a thorough discussion of the implications, but there may be minor mistakes. The data loading is clearly described. | An outstanding exploration of the data that critically discusses the implications with very little that could be improved. The data loading is perfectly described. |
| **Regression Implementation: Methodology (10 Marks)** | A completely unsatisfactory or missing implementation of a regression model methodology as per requirements: unsuitable model choices, limited/missing justifications, confused explanations, poor pre-processing and optimization schemes. | A poor attempt at implementation of a regression model methodology as per requirements: poor model choices, limited justifications, confused explanations, poor pre-processing and optimization scheme. | A good implementation a regression model methodology as per requirements: two adequate model choices, justifications, explanations, pre-processing and optimization schemes. Some mistakes or minor omissions. | A very good implementation a regression model methodology as per requirements: two very good model choices, justifications, explanations, pre-processing and optimization schemes. Very few mistakes or very minor omissions. | An excellent implementation a regression model methodology as per requirements: two excellent model choices, justifications, explanations, pre-processing and optimization schemes. Very few minor mistakes and no omissions. | An outstanding implementation of regression model methodology with very little that could be improved. Only very minor, if any, issues overall. Goes above and beyond spec requirements and contains elements beyond those covered in class. |
| **Regression Implementation: Evaluation (15 Marks)** | A completely unsatisfactory or missing evaluation of regression model trials as per requirements: unsuitable metric choices, limited/missing justifications, confused explanations, very poor optimization execution. | A poor attempt at the evaluation of the regression model trials as per requirements: poor metric choices, limited justifications, confused explanations, poor optimization execution. Overall mistakes or minor omissions. | A good attempt at the evaluation of the regression model trials as per requirements: adequate metric choices, valid justifications, relevant explanations, good optimization execution. Some mistakes or minor omissions. | A well done, complete evaluation of the regression model trials as per requirements: very good metric choices, valid, detailed justifications, relevant detailed explanations, very good optimization execution. Very few mistakes or very minor omissions. | An excellent evaluation of the regression model trials as per requirements: excellent metric choices, valid, detailed and robust justifications, relevant, detailed explanations, excellent optimization execution. Very few minor mistakes and no omissions. | An outstanding evaluation of the regression model trials with very little that could be improved. Only very minor, if any, issues overall. Goes above and beyond spec requirements and contains elements beyond those covered in class. |

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| **Regression Implementation: Critical Review (5 Marks)** | A completely unsatisfactory or missing critical review of the regression task implementation. Very little discussion, or highly irrelevant topic choice. | A poor attempt at critically reviewing of the overall regression task. Discussion missing some key points. Poor or largely incorrect coverage of approach strengths and weaknesses. Poor or irrelevant alternate approach suggestion. | A good attempt at critically reviewing of the overall regression task. Adequate, mostly relevant coverage of approach strengths and weaknesses. Mostly relevant alternate approach suggestion. | A very good attempt at critically reviewing of the overall regression task. Well crafted, relevant coverage of approach strengths and weaknesses. Well thought out and relevant alternate approach suggestion. | An excellent critical review of the overall regression task. Very well crafted, highly topical coverage of approach strengths and weaknesses. Excellently thought out and relevant alternate approach suggestion. | An outstanding critical review of the overall regression task. Excellently crafted coverage of approach strengths and weaknesses. Excellently thought-out relevant, multiple alternate approach suggestions. |
| **Classification Implementation: Feature Crafting (10 Marks)** | Feature crafting routine, resulting output labels and impact discussion missing or severely inadequate. | A poorly articulated approach to feature crafting, or use of provisional thresholding scheme (1500) with poorly discussed class balance and threshold impact. Some errors in key areas of threshold implementation. | An adequate approach to feature crafting using the provided thresholding scheme (1500). Decently discussed class balance and threshold impact. Minor errors in key areas of threshold implementation. | A very good approach to feature crafting using a simple, but custom thresholding scheme that is justified. Well discussed class balance and threshold impact. Minor errors in non-key areas of threshold implementation. | An excellent approach to feature crafting using an algorithmic, well-justified thresholding scheme, containing more than two output classes. Very well discussed class balance and threshold impact. Few minor errors in non-key areas of threshold implementation. | An outstanding approach to feature crafting using an algorithmic, well-justified thresholding scheme, containing more than two output classes. Excellently discussed class balance and threshold impact. Overall issues very minor, if present at all. |
| **Classification Implementation: Methodology (10 Marks)** | A completely unsatisfactory or missing implementation of a classification model methodology as per requirements: unsuitable model choices, limited/missing justifications, confused explanations, poor pre-processing and optimization schemes. | A poor attempt at implementation of a classification model methodology as per requirements: poor model choices, limited justifications, confused explanations, poor pre-processing and optimization scheme. | A good implementation a classification model methodology as per requirements: two adequate model choices, justifications, explanations, pre-processing and optimization schemes. Some mistakes or minor omissions. | A very good implementation a classification model methodology as per requirements: two very good model choices, justifications, explanations, pre-processing and optimization schemes. Very few mistakes or very minor omissions. | An excellent implementation a classification model methodology as per requirements: two excellent model choices, justifications, explanations, pre-processing and optimization schemes. Very few minor mistakes and no omissions. | An outstanding implementation of classification model methodology with very little that could be improved. Only very minor, if any, issues overall. Goes above and beyond spec requirements and contains elements beyond those covered in class. |

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| **Classification Implementation: Evaluation (15 Marks)** | A completely unsatisfactory or missing evaluation of classification model trials as per requirements: unsuitable metric choices, limited/missing justifications, confused explanations, very poor optimization execution. | A poor attempt at the evaluation of the classification model trials as per requirements: poor metric choices, limited justifications, confused explanations, poor optimization execution. Overall mistakes or minor omissions. | A good attempt at the evaluation of the classification model trials as per requirements: adequate metric choices, valid justifications, relevant explanations, good optimization execution. Some mistakes or minor omissions. | A well done, complete evaluation of the classification model trials as per requirements: very good metric choices, valid, detailed justifications, relevant detailed explanations, very good optimization execution. Very few mistakes or very minor omissions. | An excellent evaluation of the classification model trials as per requirements: excellent metric choices, valid, detailed and robust justifications, relevant, detailed explanations, excellent optimization execution. Very few minor mistakes and no omissions. | An outstanding evaluation of the classification model trials with very little that could be improved. Only very minor, if any, issues overall. Goes above and beyond spec requirements and contains elements beyond those covered in class. |
| **Classification Implementation: Critical Review (5 Marks)** | A completely unsatisfactory or missing critical review of the classification task implementation. Very little discussion, or highly irrelevant topic choice. | A poor attempt at critically reviewing of the overall classification task. Discussion missing some key points. Poor or largely incorrect coverage of approach strengths and weaknesses. Poor or irrelevant alternate approach suggestion. | A good attempt at critically reviewing of the overall classification task. Adequate, mostly relevant coverage of approach strengths and weaknesses. Mostly relevant alternate approach suggestion. | A very good attempt at critically reviewing of the overall classification task. Well crafted, relevant coverage of approach strengths and weaknesses. Well thought out and relevant alternate approach suggestion. | An excellent critical review of the overall classification task. Very well crafted, highly topical coverage of approach strengths and weaknesses. Excellently thought out and relevant alternate approach suggestion. | An outstanding critical review of the overall classification task. Excellently crafted coverage of approach strengths and weaknesses. Excellently thought-out relevant, multiple alternate approach suggestions. |
| **Conclusions (5 Marks)** | No conclusion or final recommendation given. | Conclusion is very superficial or generic when discussing tasks. Recommendations not congruent with experimentation results. | A good conclusion that discusses some relevant details of the work done, though some points may be generic. Recommendations congruent with experimentation results | A good conclusion that discusses many relevant details of the work done, though some points may be generic. Recommendations very much in-line with experimentation results. | An excellent conclusion that discusses many specific details of the work done. Recommendations very much in-line with most experimentation results. | An outstanding conclusion that discusses relevant and specific details of all tasks. Recommendations very much in-line with all experimentation results. |
| **Report Presentation and language (10 marks)** | A completely unsatisfactory report layout and language, often being disjointed and confusing to read. | Poor language and layout overall, with many mistakes in the report layout and language that is often hard to follow, | Good layout and language overall, though many minor mistakes in report layout and quality of language | Very good layout and language. Could be improved with better grammar and minor fixes to presentation mistakes. | Excellent language and layout, minor issues with grammar or presentation. | Outstanding layout and language used throughout. Report presentation follows guidelines perfectly and language clear and professional throughout. |

**If you are unsure about any of these instructions, then please get in touch with your module leader (ideally during lab of open office hours.)**